



NSeqArpKeys

PATTERN INSTRUMENT

Turn integer sequences into melodies and chords through Forte pitch-class sets.



Each pattern integer is a binary note selector: one set bit chooses one pitch, and several set bits make a chord. Change the Forte set to remap the same sequence to a new pitch collection. Assign a different pattern and set to each MIDI trigger key.

Create more material with [Sequence Operator](#), [Rhythm Navigator](#), and [KComplexExplorer](#).

1. How integer patterns become notes

A short sequence of numbers becomes a melody, chords, and rests when you choose a Forte set.

In **Melodic** mode, each space-separated integer in **Pattern** is one step. The chosen **Forte Set** supplies pitch classes (0 = C through 11 = B); NSeqArpKeys repeats them in ascending order across octaves. The **Octave** control chooses the starting position, called the anchor below.

Read the magnitude of a step in binary from right to left. The rightmost bit, bit 0, selects the anchor note; bit 1 selects the next note in the expanded set; bit 2 selects the note after that. Several set bits play together. A positive number walks upward, a negative number walks downward from the same anchor, and zero starts no note.

Hear the mapping

Choose 7-35.11, the diatonic set [0 2 4 5 7 9 11] (C D E F G A B), and set Octave to 4. The anchor is C (MIDI 48), followed upward by D (50), E (52), and F (53).

Step	Binary magnitude	Notes
1	001	C
2	010	D
4	100	E
3	011	C + D
5	101	C + E
7	111	C + D + E
0	000	Rest
-2	010	B below the anchor (MIDI 47)

Enter 1 2 4 8 3 5 7 0 to hear single notes become chords. Keep the sequence and choose another Forte Set to hear new pitch material with the same bit relationships. The MIDI key you press triggers its assigned pattern; it does not transpose the output. **Transpose** moves melodic notes by positions in the selected Forte set: with the diatonic set, +1 moves C to D and +7 moves C up an octave. Negative values move down; notes outside MIDI 0-127 are omitted. In Rhythmic mode, Transpose moves drum notes by semitones. The appendix lists every base Forte set, its pitch classes, and its interval vector.

2. Get started

Load NSeqArpKeys as an instrument plug-in in a compatible DAW.

Install

Plug-in

Copy the complete `NSeqArpKeys.vst3` bundle to your VST3 folder and rescan in your DAW. The macOS archive also includes `NSeqArpKeys.component` for Audio Units.

Keep plug-in bundles intact. VST3 folders: Windows `C:\Program Files\Common Files\VST3` , Linux `~/vst3` , macOS `~/Library/Audio/Plug-Ins/VST3` . For Audio Units on macOS, use `~/Library/Audio/Plug-Ins/Components` .

Make your first pattern

- 1 Open **Presets**, choose **Single Note Pulse**, and click **Load**. This assigns a playable example to C4 (MIDI 60).
- 2 Hold C4 on the on-screen keyboard or send MIDI note 60 to the plug-in. The pattern plays while that key is held.
- 3 Release C4 to stop. To keep it running while editing, turn on **Latch (keep playing)**, trigger C4 once, and release it.
- 4 Click the same key again, or use **Stop Key** or **Stop All**, to stop a latched pattern.

A melodic key needs both a nonzero Pattern step and a Forte Set. Rhythmic mode uses Drum notes instead. The Init preset is deliberately silent.

Make patterns with companion apps

- **Sequence Operator** (necg777.github.io/sequence-operator/) creates and transforms integer sequences. Copy its result into a Melodic **Pattern** to hear those numbers as notes and chords.
- **Rhythm Navigator** (necg777.github.io/rhythm-navigator/) supplies onset patterns. Put the Sequence Operator values at those onsets and `0` at rests. For drums, use **Combined sequence** → **Copy** in its drum sequencer and paste the decimal masks into a Rhythmic **Pattern**. Use at most 16 lanes, set **Velocity bits/lane** to `1`, enter Drum notes in its displayed bit order, and set **Steps/QN** to its displayed denominator when that is 16 or less.
- **KComplexExplorer** (necg777.github.io/KComplexExplorer/) lets you audition and compare pitch-class sets, inspect their Forte numbers and interval vectors, and generate harmonic matrices. Choose a set in **Find Set / Forte Set**; different trigger keys can use different sets. Its padded suffix `.09` appears as `.9` in NSeqArpKeys.

Use KComplexExplorer for the pitch collection, Sequence Operator for the note-mask stream, and Rhythm Navigator for its timing; then trigger the finished pattern from a MIDI key in NSeqArpKeys.

3. Work in the editor

The compact editor opens at up to 820 × 600 logical pixels. Drag its bottom-right corner to resize down to 640 × 400; use the scrollbars to reach all controls in smaller windows. Preset navigation stays at the top, and switching modes or browsers keeps your window size.

The keyboard selects a trigger note. The controls below it edit the assignment for that key. Each of the 128 MIDI keys can have its own pattern, set, channel, octave, gate, subdivision and expression.

Control	What it does
Preset name / < / >	Shows the loaded preset and steps through the preset library. An asterisk means the current setup has changed.
Presets / Pattern Bank / Save	Opens presets or the built-in Factory Patterns library. Save updates a user preset or opens a new copy.
Keyboard / Playing	Triggers a pattern and selects the MIDI key whose assignment is shown. The label below lists running keys. Right-click the keyboard to assign the selected bank pattern.
Latch / Stop Key / Stop All	Keeps patterns running after release, stops the selected key, or stops every active key.
Preview sound	Uncheck to silence the built-in tones while generated MIDI continues.
Meter Num / Global Steps/QN	Global grouping and default subdivision. Steps/QN controls the duration of one step at the current tempo.
Channel / Octave	Sets the selected key's MIDI output channel and pitch region.
Gate / Fixed Steps	Adds a gated fraction of the span through rests and a fixed duration in steps.
Pattern (length) / step preview	Space-separated step values. The label shows the number of items; the preview shows the first 16 steps. Zero creates a rest and extends the previous sounding step's gate span.
Name / Mode / Steps/QN	Name one pattern; choose melodic or rhythmic interpretation and its step subdivision. Zero inherits the global rate.
Velocity / Transpose / Rotate / Reverse	Shape output strength, move melodic pitches through the selected Forte set (or rhythmic drum pitches by semitones), and change the integer step order.
Drum notes / Velocity bits/lane	In rhythmic mode enter 1-16 MIDI pitches. One numeric setting chooses the same bit width (1-7) for every pitch.
Colour / Tags / Favourite	Organize patterns visually and make them easy to find in the bank.
Cut / Clear / Copy / Paste	Remove or copy an assignment. Paste all settings, just sequence, timing, or expression.
Duplicate / Make independent	Duplicate to the next key as a separate copy, or break a shared pattern link.
Save pattern / Load pattern	Store or recall a single assignment as a <code>.nseqpattern</code> file.
Import bank / Export bank	Merge a whole pattern bank from a <code>.nseqbank</code> XML file, or save the whole bank to one file.
MIDI range / Transpose by key / Assign range	Copy an assignment into the inclusive range, optionally adding one Forte-set degree per key in Melodic mode or one semitone per key in Rhythmic mode.
Find Set / Forte Set	Search by Forte ID, common name, or pitch classes, then choose a matching pitch-class set.

Integer fields accept up to 4096 space-separated values, with a 45,056-character input limit. Names, tags and preset details also have length limits. Invalid input is outlined in red and is not applied. Blank Pattern clears the sequence.

Change a running pattern

With Latch on, trigger and release a key, then edit its controls while listening. Most edits restart that key on the next audio block, preserving trigger velocity. Changes to tempo or Steps/QN preserve musical phase and pending note-offs.

Stopping playback. Stop Key ends the selected key. Stop All, a DAW transport stop, or MIDI All Notes Off / All Sound Off ends every pattern, including latched keys and auditions.

4. Patterns, pitch, and timing

Pattern values

Enter whole numbers separated by spaces. Each number is one step. Positive numbers are bit masks over ascending notes of the selected Forte set. Negative numbers use the bits of their absolute value in the opposite direction from the Octave anchor. The least significant bit selects the anchor in both directions. Zero starts no new note.

Example - four steps

1 2 3 0

1 plays the first available note, 2 the next, and 3 plays both together. 0 creates a rest after the chord.

Octave moves the base position used by these bit masks. Channel sets the outgoing MIDI channel from 1 to 16. Notes outside MIDI's 0-127 range are ignored.

Gate across zero steps

A sounding step owns the zero steps that follow it, including zero steps at the end of a looping pattern. Gate is a percentage of that entire span, from the note onset to the next sounding step.

Example - a three-step span

1 0 0 2

The first note has three steps available: its own step and two following zeros. At 50% Gate it sounds for 1.5 steps; at 100% it reaches the next sounding step. A zero step does not retrigger the note.

Gate ranges from 0% to 200%. Values above 100% can overlap later notes. Fixed Steps ranges from 0 to 16 in increments of 0.01. The two lengths add: $\text{note length} = \text{Fixed Steps} + \text{Gate} \times \text{sounding span}$. For example, a three-step span with Gate 50% and Fixed Steps 0.5 lasts 2 steps. Releasing the trigger key stops the pattern immediately when Latch is off.

Tempo and meter

Pattern steps follow the DAW tempo when a host supplies one. Otherwise the instrument uses 120 BPM. One step lasts $60 / (\text{BPM} \times \text{effective Steps/QN})$ seconds: at 120 BPM and Steps/QN 4, a step lasts 0.125 seconds. A per-key value of 0 inherits the global setting. Meter Num is saved but does not currently change step duration or pattern length.

Rhythmic mode and variation

Rhythmic mode needs no Forte set. Enter 1-16 Drum notes; this sets the active lane count. With one velocity bit for each lane, bit 0 plays the first pitch, bit 1 the second, and so on. Try 1 0 0 0 2 0 0 0 for notes 36 and 38 with the default map. Set Channel to match the receiving instrument; changing mode preserves it.

Velocity bits/lane is one number from 1-7, shared by every active drum lane as in GateRunner. Lane 1 occupies the least significant bits of a positive decimal step mask; each later lane uses the next group of bits. A lane value of zero rests, while nonzero values scale from quiet to velocity 127. For example, two lanes with two bits each decode step 9 as level 1/3 on lane 1 and 2/3 on lane 2. Positive masks support all 16 lanes at seven bits each (up to 112 bits). Legacy signed 32-bit masks remain accepted. Base velocity and incoming key velocity further scale the result. Rotate and Reverse reorder the integer steps.

5. Forte sets

A Forte set supplies the pitch classes from which Pattern values choose notes. Each menu match shows its Forte ID, any common name, and its pitch classes in brackets. The number before the dot identifies the set class; the suffix `.0` through `.11` transposes its listed pitch classes upward by that many semitones modulo 12. The interval vector stays the same under transposition. The appendix gives every base set and the rule for reading its transpositions.

1. Select the MIDI key you want to configure.
2. Type part of an ID, a common name such as `diatonic`, or pitch classes in **Find Set**.
3. Open **Forte Set** and choose a match. The selected set is shown underneath.

Example

Base 7-35.0 is `[0 1 3 5 6 8 10]`. Adding 11 semitones modulo 12 gives 7-35.11 - diatonic scale `[0 2 4 5 7 9 11]`. With this set, `1 2 4 8` walks upward through C, D, E, F at the selected Octave anchor.

Choose **(No set)** to clear the selected key's set. A melodic pattern without a set is silent; rhythmic mode uses Drum notes independently. Changing search text only filters the menu; choose a result to change the assignment.

Live performance

Normally, a pattern runs only while its trigger note is held. Latch changes note release behavior: the first trigger keeps the pattern running, and another trigger of the same key switches it off. Turning Latch off stops patterns whose trigger keys are no longer held.

Turn off **Preview sound** to silence the built-in tones while MIDI continues. This setting is saved with the setup. Route the plug-in's MIDI output to another instrument in your DAW and match Channel to the receiving instrument. MIDI routing depends on the host.

6. Presets

A preset captures the meter, latch setting, selected key, and all 128 key assignments. The library has six factory examples and can hold your own presets.

Browse and load

Open **Presets**. Search names, categories, tags, and descriptions; narrow results with the Category menu or **Favourites only**. Select a row to inspect its details, then click **Load**. Loading replaces the current setup and stops any active patterns.

Save and manage

Action	Result
Save New	Stores the current setup under the name, category, tags, and description shown on the right.
Save	Updates the currently loaded user preset. For a factory preset, opens the browser with a suggested copy name.
Update	Writes the current setup and edited details to the selected user preset. Factory presets cannot be updated.
Duplicate	Creates and loads a user copy of the selected preset.
Favourite / Unfavourite	Adds or removes the selected preset from the favourites filter.
Delete	After confirmation, removes a user preset from disk. Factory presets cannot be deleted.
Import / Export	Copies a <code>.nseqpreset</code> file into the library, or saves the selected library preset to a file. Import selects the new row; click Load to apply it.

Where presets live: in the OS application-data folder under `NSeqArpKeys/Presets` (for example, `%APPDATA%\NSeqArpKeys\Presets` on Windows). The plug-in formats share this folder on each OS.

Pattern files and whole-bank transfer

Save pattern and Load pattern use the separate `NSeqArpKeys/Patterns` folder. Pattern Bank searches names and tags, filters favourites, edits colour and details, auditions a selected pattern, and assigns it as an independent copy or shared link. Import bank and Export bank transfer all bank entries in one `.nseqbank` XML file. Import updates local entries with matching IDs, adds new entries, and retains other local entries. An entry without an ID receives one on import. Export includes all entries even when a search or favourites filter is active. Linked keys follow edits to any member; Make independent breaks the link on the selected key. Duplicate and range assignment always make independent copies. Whole presets embed shared definitions and key assignments, so they reopen without the local bank. Loading one pattern affects only the selected key. Undo/Redo lasts for the editor session and resets when loading a whole preset or project.

7. If something is not working

What you notice	What to check
No notes when a key is held	Enter a nonzero Pattern value. Melodic mode needs a Forte Set; rhythmic mode needs Drum notes and a nonzero velocity level for an active lane. Check the receiving MIDI channel.
The pattern stops when you release the key	That is the default behavior. Turn on Latch before triggering the key if you want it to continue during edits.
A pattern keeps playing	Click its trigger key again while Latch is on, use Stop Key for the selected key, or use Stop All.
VST3 does not appear in the DAW	Keep the outer <code>NSeqArpKeys.vst3</code> folder intact in the VST3 directory and rescan plug-ins in the DAW.
A preset does not appear	Check the browser's search, category, and Favourites only filters. User files must be valid <code>.nseqpreset</code> files in the preset folder.

Quick reference

To hear a pattern

Assign Pattern + Forte Set, then hold the trigger note. Use Latch to keep it running while you edit.

To share a setup

Save New, select the preset in the browser, then Export. Send the resulting `.nseqpreset` file.

NSeqArpKeys 1.5.1 | Windows, Linux, macOS | VST3; Audio Units on macOS.

8. Create variations

Choose a source key in the editor, then click **Generate variations**. This page takes a snapshot of that assignment. Changing its controls and auditioning results leaves your assigned patterns unchanged.

Polynomial mappings

For a sequence of length N , positions start at zero and the mapping is $p(i) = a i^3 + b i^2 + c i + d \pmod{N}$. Output step i reads source step $p(i)$. The coefficients are labelled Cubic, Quadratic, Linear and Constant. Quadratic, Cubic and Rotation presets supply starting values; edit any coefficient to make a Custom polynomial.

Every polynomial is allowed. A **permutation** uses every source position exactly once. Other mappings repeat some source steps and omit others, allowing repetition, thinning and collapse. **Inverse permutation** is enabled only for permutations; changing to a mapping with repeated positions clears the inverse selection.

For example, $p(i) = 2i \pmod{8}$ maps 1 2 3 4 5 6 7 8 to 1 3 5 7 1 3 5 7, then 1 5 1 5 1 5 1 5, then all ones.

Keep zero steps in place maps only nonzero positions, using their count as the modulus, and leaves zeros fixed. It preserves the integer rhythm; whether a nonzero mask sounds depends on its mode, pitches and MIDI bounds.

Vertical flip

Find the lowest and highest occupied bit positions across the whole melodic sequence. Every occupied position k maps to $\min + \max - k$. Thus 2 6 8 becomes 8 12 2. Zero steps stay zero, each step retains its set-bit count, and negative values retain their downward direction. A reflection that exceeds the melodic signed 32-bit range is rejected.

In Rhythmic mode the same reflection moves occupied drum lanes. Each lane's complete velocity word moves intact, including masks wider than 32 bits. Legacy negative rhythmic masks are represented as equivalent unsigned masks after a flip.

Repeat across keys

First key and **Number of keys** choose consecutive MIDI keys. **Starting application** chooses the first power: zero is the original, one applies the operation once. **Applications between keys** is the stride. With a start of zero and stride two, keys receive applications 0, 2, 4, 6 and so on.

Each application composes the operation with the previous result. The list identifies repeated patterns. Permutations repeat from application zero; other mappings can pass through a transient before settling into a cycle. The status reports the application from which the mapping is periodic and its cycle length. Identical source values can make patterns repeat sooner.

Compare, audition and save

Select a variation in the list. The two grids compare its bits or drum lanes with the original. Columns are playback steps; melodic rows are magnitude bit positions, with negative-direction steps shown in purple. Drum-row brightness follows the encoded velocity. The grids include the source's existing rotation and reversal. Use **First visible step** to inspect long sequences, and the grid scrollbar to reach distant bit positions.

Audition original and **Audition variation** replace the current audition on the source key without changing its saved assignment. A playhead follows actual playback and pages through longer loops. **Stop audition**, leaving this page, closing the editor, transport stop and MIDI panic stop audition playback. The existing Preview sound setting still controls the built-in sound; MIDI continues to the receiving instrument.

Assign variations writes independent copies to the chosen keys, retaining Forte, channel, timing, transposition and expression settings. One Undo restores the entire batch, including prior linked assignments. **Save family to bank** saves every listed variation under the Family name and Tags, with application counts in the names. Families appear in the user Pattern Bank and use ordinary portable pattern files. Reopen this page after a project restore to capture the new source.

Appendix A. Forte pitch classes and interval vectors

This table lists all 352 base IDs in NSeqArpKeys's Forte catalogue, including the empty set. Pitch classes use C = 0, C#/D♭ = 1, D = 2, D#/E♭ = 3, E = 4, F = 5, F#/G♭ = 6, G = 7, G#/A♭ = 8, A = 9, A#/B♭ = 10, and B = 11. The displayed pitch classes belong to the .0 form of each ID.

For any other suffix .n, add n to each listed pitch class modulo 12, then sort the results. For example, 7-35.0 is [0 1 3 5 6 8 10]; 7-35.11 is [0 2 4 5 7 9 11]. Symmetric sets can have fewer than 12 distinct transpositions, so the selector only shows distinct forms. The empty set supplies no melodic notes.

The interval vector <1 2 3 4 5 6> counts unordered pitch-class pairs at each interval class in semitones, folding intervals larger than six back toward zero. For instance, [0 4 7] has one pair each at interval classes 3, 4, and 5, so its vector is <0 0 1 1 1 0>. Transposition leaves the vector unchanged. The table is generated from the plug-in's ForteCSV.cpp catalogue.

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
0-1.0	[]	<0 0 0 0 0 0>
1-1.0	[0]	<0 0 0 0 0 0>
2-1.0	[0 1]	<1 0 0 0 0 0>
2-2.0	[0 2]	<0 1 0 0 0 0>
2-3.0	[0 3]	<0 0 1 0 0 0>
2-4.0	[0 4]	<0 0 0 1 0 0>
2-5.0	[0 5]	<0 0 0 0 1 0>
2-6.0	[0 6]	<0 0 0 0 0 1>
3-1.0	[0 1 2]	<2 1 0 0 0 0>
3-2A.0	[0 1 3]	<1 1 1 0 0 0>
3-2B.0	[0 2 3]	<1 1 1 0 0 0>
3-3A.0	[0 1 4]	<1 0 1 1 0 0>
3-3B.0	[0 3 4]	<1 0 1 1 0 0>
3-4A.0	[0 1 5]	<1 0 0 1 1 0>
3-4B.0	[0 4 5]	<1 0 0 1 1 0>
3-5A.0	[0 1 6]	<1 0 0 0 1 1>
3-5B.0	[0 5 6]	<1 0 0 0 1 1>
3-6.0	[0 2 4]	<0 2 0 1 0 0>
3-7A.0	[0 2 5]	<0 1 1 0 1 0>
3-7B.0	[0 3 5]	<0 1 1 0 1 0>
3-8A.0	[0 2 6]	<0 1 0 1 0 1>
3-8B.0	[0 4 6]	<0 1 0 1 0 1>
3-9.0	[0 2 7]	<0 1 0 0 2 0>
3-10.0	[0 3 6]	<0 0 2 0 0 1>
3-11A.0	[0 3 7]	<0 0 1 1 1 0>
3-11B.0	[0 4 7]	<0 0 1 1 1 0>
3-12.0	[0 4 8]	<0 0 0 3 0 0>
4-1.0	[0 1 2 3]	<3 2 1 0 0 0>
4-2A.0	[0 1 2 4]	<2 2 1 1 0 0>
4-2B.0	[0 2 3 4]	<2 2 1 1 0 0>
4-3.0	[0 1 3 4]	<2 1 2 1 0 0>
4-4A.0	[0 1 2 5]	<2 1 1 1 1 0>
4-4B.0	[0 3 4 5]	<2 1 1 1 1 0>
4-5A.0	[0 1 2 6]	<2 1 0 1 1 1>
4-5B.0	[0 4 5 6]	<2 1 0 1 1 1>
4-6.0	[0 1 2 7]	<2 1 0 0 2 1>
4-7.0	[0 1 4 5]	<2 0 1 2 1 0>
4-8.0	[0 1 5 6]	<2 0 0 1 2 1>
4-9.0	[0 1 6 7]	<2 0 0 0 2 2>
4-10.0	[0 2 3 5]	<1 2 2 0 1 0>
4-11A.0	[0 1 3 5]	<1 2 1 1 1 0>
4-11B.0	[0 2 4 5]	<1 2 1 1 1 0>

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
4-12A.0	[0 2 3 6]	<1 1 2 1 0 1>
4-12B.0	[0 3 4 6]	<1 1 2 1 0 1>
4-13A.0	[0 1 3 6]	<1 1 2 0 1 1>
4-13B.0	[0 3 5 6]	<1 1 2 0 1 1>
4-14A.0	[0 2 3 7]	<1 1 1 1 2 0>
4-14B.0	[0 4 5 7]	<1 1 1 1 2 0>
4-z15A.0	[0 1 4 6]	<1 1 1 1 1 1>
4-z15B.0	[0 2 5 6]	<1 1 1 1 1 1>
4-16A.0	[0 1 5 7]	<1 1 0 1 2 1>
4-16B.0	[0 2 6 7]	<1 1 0 1 2 1>
4-17.0	[0 3 4 7]	<1 0 2 2 1 0>
4-18A.0	[0 1 4 7]	<1 0 2 1 1 1>
4-18B.0	[0 3 6 7]	<1 0 2 1 1 1>
4-19A.0	[0 1 4 8]	<1 0 1 3 1 0>
4-19B.0	[0 3 4 8]	<1 0 1 3 1 0>
4-20.0	[0 1 5 8]	<1 0 1 2 2 0>
4-21.0	[0 2 4 6]	<0 3 0 2 0 1>
4-22A.0	[0 2 4 7]	<0 2 1 1 2 0>
4-22B.0	[0 3 5 7]	<0 2 1 1 2 0>
4-23.0	[0 2 5 7]	<0 2 1 0 3 0>
4-24.0	[0 2 4 8]	<0 2 0 3 0 1>
4-25.0	[0 2 6 8]	<0 2 0 2 0 2>
4-26.0	[0 3 5 8]	<0 1 2 1 2 0>
4-27A.0	[0 2 5 8]	<0 1 2 1 1 1>
4-27B.0	[0 3 6 8]	<0 1 2 1 1 1>
4-28.0	[0 3 6 9]	<0 0 4 0 0 2>
4-z29A.0	[0 1 3 7]	<1 1 1 1 1 1>
4-z29B.0	[0 4 6 7]	<1 1 1 1 1 1>
5-1.0	[0 1 2 3 4]	<4 3 2 1 0 0>
5-2A.0	[0 1 2 3 5]	<3 3 2 1 1 0>
5-2B.0	[0 2 3 4 5]	<3 3 2 1 1 0>
5-3A.0	[0 1 2 4 5]	<3 2 2 2 1 0>
5-3B.0	[0 1 3 4 5]	<3 2 2 2 1 0>
5-4A.0	[0 1 2 3 6]	<3 2 2 1 1 1>
5-4B.0	[0 3 4 5 6]	<3 2 2 1 1 1>
5-5A.0	[0 1 2 3 7]	<3 2 1 1 2 1>
5-5B.0	[0 4 5 6 7]	<3 2 1 1 2 1>
5-6A.0	[0 1 2 5 6]	<3 1 1 2 2 1>
5-6B.0	[0 1 4 5 6]	<3 1 1 2 2 1>
5-7A.0	[0 1 2 6 7]	<3 1 0 1 3 2>
5-7B.0	[0 1 5 6 7]	<3 1 0 1 3 2>
5-8.0	[0 2 3 4 6]	<2 3 2 2 0 1>
5-9A.0	[0 1 2 4 6]	<2 3 1 2 1 1>
5-9B.0	[0 2 4 5 6]	<2 3 1 2 1 1>
5-10A.0	[0 1 3 4 6]	<2 2 3 1 1 1>
5-10B.0	[0 2 3 5 6]	<2 2 3 1 1 1>
5-11A.0	[0 2 3 4 7]	<2 2 2 2 2 0>
5-11B.0	[0 3 4 5 7]	<2 2 2 2 2 0>
5-z12.0	[0 1 3 5 6]	<2 2 2 1 2 1>
5-13A.0	[0 1 2 4 8]	<2 2 1 3 1 1>
5-13B.0	[0 2 3 4 8]	<2 2 1 3 1 1>
5-14A.0	[0 1 2 5 7]	<2 2 1 1 3 1>
5-14B.0	[0 2 5 6 7]	<2 2 1 1 3 1>
5-15.0	[0 1 2 6 8]	<2 2 0 2 2 2>
5-16A.0	[0 1 3 4 7]	<2 1 3 2 1 1>
5-16B.0	[0 3 4 6 7]	<2 1 3 2 1 1>
5-z17.0	[0 1 3 4 8]	<2 1 2 3 2 0>

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
5-z18A.0	[0 1 4 5 7]	<2 1 2 2 2 1>
5-z18B.0	[0 2 3 6 7]	<2 1 2 2 2 1>
5-19A.0	[0 1 3 6 7]	<2 1 2 1 2 2>
5-19B.0	[0 1 4 6 7]	<2 1 2 1 2 2>
5-20A.0	[0 1 5 6 8]	<2 1 1 2 3 1>
5-20B.0	[0 2 3 7 8]	<2 1 1 2 3 1>
5-21A.0	[0 1 4 5 8]	<2 0 2 4 2 0>
5-21B.0	[0 3 4 7 8]	<2 0 2 4 2 0>
5-22.0	[0 1 4 7 8]	<2 0 2 3 2 1>
5-23A.0	[0 2 3 5 7]	<1 3 2 1 3 0>
5-23B.0	[0 2 4 5 7]	<1 3 2 1 3 0>
5-24A.0	[0 1 3 5 7]	<1 3 1 2 2 1>
5-24B.0	[0 2 4 6 7]	<1 3 1 2 2 1>
5-25A.0	[0 2 3 5 8]	<1 2 3 1 2 1>
5-25B.0	[0 3 5 6 8]	<1 2 3 1 2 1>
5-26A.0	[0 2 4 5 8]	<1 2 2 3 1 1>
5-26B.0	[0 3 4 6 8]	<1 2 2 3 1 1>
5-27A.0	[0 1 3 5 8]	<1 2 2 2 3 0>
5-27B.0	[0 3 5 7 8]	<1 2 2 2 3 0>
5-28A.0	[0 2 3 6 8]	<1 2 2 2 1 2>
5-28B.0	[0 2 5 6 8]	<1 2 2 2 1 2>
5-29A.0	[0 1 3 6 8]	<1 2 2 1 3 1>
5-29B.0	[0 2 5 7 8]	<1 2 2 1 3 1>
5-30A.0	[0 1 4 6 8]	<1 2 1 3 2 1>
5-30B.0	[0 2 4 7 8]	<1 2 1 3 2 1>
5-31A.0	[0 1 3 6 9]	<1 1 4 1 1 2>
5-31B.0	[0 2 3 6 9]	<1 1 4 1 1 2>
5-32A.0	[0 1 4 6 9]	<1 1 3 2 2 1>
5-32B.0	[0 3 5 8 9]	<1 1 3 2 2 1>
5-33.0	[0 2 4 6 8]	<0 4 0 4 0 2>
5-34.0	[0 2 4 6 9]	<0 3 2 2 2 1>
5-35.0	[0 2 4 7 9]	<0 3 2 1 4 0>
5-z36A.0	[0 1 2 4 7]	<2 2 2 1 2 1>
5-z36B.0	[0 3 5 6 7]	<2 2 2 1 2 1>
5-z37.0	[0 3 4 5 8]	<2 1 2 3 2 0>
5-z38A.0	[0 1 2 5 8]	<2 1 2 2 2 1>
5-z38B.0	[0 3 6 7 8]	<2 1 2 2 2 1>
6-1.0	[0 1 2 3 4 5]	<5 4 3 2 1 0>
6-2A.0	[0 1 2 3 4 6]	<4 4 3 2 1 1>
6-2B.0	[0 2 3 4 5 6]	<4 4 3 2 1 1>
6-z3A.0	[0 1 2 3 5 6]	<4 3 3 2 2 1>
6-z3B.0	[0 1 3 4 5 6]	<4 3 3 2 2 1>
6-z4.0	[0 1 2 4 5 6]	<4 3 2 3 2 1>
6-5A.0	[0 1 2 3 6 7]	<4 2 2 2 3 2>
6-5B.0	[0 1 4 5 6 7]	<4 2 2 2 3 2>
6-z6.0	[0 1 2 5 6 7]	<4 2 1 2 4 2>
6-7.0	[0 1 2 6 7 8]	<4 2 0 2 4 3>
6-8.0	[0 2 3 4 5 7]	<3 4 3 2 3 0>
6-9A.0	[0 1 2 3 5 7]	<3 4 2 2 3 1>
6-9B.0	[0 2 4 5 6 7]	<3 4 2 2 3 1>
6-z10A.0	[0 1 3 4 5 7]	<3 3 3 3 2 1>
6-z10B.0	[0 2 3 4 6 7]	<3 3 3 3 2 1>
6-z11A.0	[0 1 2 4 5 7]	<3 3 3 2 3 1>
6-z11B.0	[0 2 3 5 6 7]	<3 3 3 2 3 1>
6-z12A.0	[0 1 2 4 6 7]	<3 3 2 2 3 2>
6-z12B.0	[0 1 3 5 6 7]	<3 3 2 2 3 2>
6-z13.0	[0 1 3 4 6 7]	<3 2 4 2 2 2>

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
6-14A.0	[0 1 3 4 5 8]	<3 2 3 4 3 0>
6-14B.0	[0 3 4 5 7 8]	<3 2 3 4 3 0>
6-15A.0	[0 1 2 4 5 8]	<3 2 3 4 2 1>
6-15B.0	[0 3 4 6 7 8]	<3 2 3 4 2 1>
6-16A.0	[0 1 4 5 6 8]	<3 2 2 4 3 1>
6-16B.0	[0 2 3 4 7 8]	<3 2 2 4 3 1>
6-z17A.0	[0 1 2 4 7 8]	<3 2 2 3 3 2>
6-z17B.0	[0 1 4 6 7 8]	<3 2 2 3 3 2>
6-18A.0	[0 1 2 5 7 8]	<3 2 2 2 4 2>
6-18B.0	[0 1 3 6 7 8]	<3 2 2 2 4 2>
6-z19A.0	[0 1 3 4 7 8]	<3 1 3 4 3 1>
6-z19B.0	[0 1 4 5 7 8]	<3 1 3 4 3 1>
6-20.0	[0 1 4 5 8 9]	<3 0 3 6 3 0>
6-21A.0	[0 2 3 4 6 8]	<2 4 2 4 1 2>
6-21B.0	[0 2 4 5 6 8]	<2 4 2 4 1 2>
6-22A.0	[0 1 2 4 6 8]	<2 4 1 4 2 2>
6-22B.0	[0 2 4 6 7 8]	<2 4 1 4 2 2>
6-z23.0	[0 2 3 5 6 8]	<2 3 4 2 2 2>
6-z24A.0	[0 1 3 4 6 8]	<2 3 3 3 3 1>
6-z24B.0	[0 2 4 5 7 8]	<2 3 3 3 3 1>
6-z25A.0	[0 1 3 5 6 8]	<2 3 3 2 4 1>
6-z25B.0	[0 2 3 5 7 8]	<2 3 3 2 4 1>
6-z26.0	[0 1 3 5 7 8]	<2 3 2 3 4 1>
6-27A.0	[0 1 3 4 6 9]	<2 2 5 2 2 2>
6-27B.0	[0 3 5 6 8 9]	<2 2 5 2 2 2>
6-z28.0	[0 1 3 4 7 9]	<2 2 4 3 2 2>
6-z29.0	[0 2 3 6 7 9]	<2 2 4 2 3 2>
6-30A.0	[0 1 3 6 7 9]	<2 2 4 2 2 3>
6-30B.0	[0 2 3 6 8 9]	<2 2 4 2 2 3>
6-31A.0	[0 1 4 5 7 9]	<2 2 3 4 3 1>
6-31B.0	[0 2 4 5 8 9]	<2 2 3 4 3 1>
6-32.0	[0 2 4 5 7 9]	<1 4 3 2 5 0>
6-33A.0	[0 2 3 5 7 9]	<1 4 3 2 4 1>
6-33B.0	[0 2 4 6 7 9]	<1 4 3 2 4 1>
6-34A.0	[0 1 3 5 7 9]	<1 4 2 4 2 2>
6-34B.0	[0 2 4 6 8 9]	<1 4 2 4 2 2>
6-35.0	[0 2 4 6 8 10]	<0 6 0 6 0 3>
6-z36A.0	[0 1 2 3 4 7]	<4 3 3 2 2 1>
6-z36B.0	[0 3 4 5 6 7]	<4 3 3 2 2 1>
6-z37.0	[0 1 2 3 4 8]	<4 3 2 3 2 1>
6-z38.0	[0 1 2 3 7 8]	<4 2 1 2 4 2>
6-z39A.0	[0 2 3 4 5 8]	<3 3 3 3 2 1>
6-z39B.0	[0 3 4 5 6 8]	<3 3 3 3 2 1>
6-z40A.0	[0 1 2 3 5 8]	<3 3 3 2 3 1>
6-z40B.0	[0 3 5 6 7 8]	<3 3 3 2 3 1>
6-z41A.0	[0 1 2 3 6 8]	<3 3 2 2 3 2>
6-z41B.0	[0 2 5 6 7 8]	<3 3 2 2 3 2>
6-z42.0	[0 1 2 3 6 9]	<3 2 4 2 2 2>
6-z43A.0	[0 1 2 5 6 8]	<3 2 2 3 3 2>
6-z43B.0	[0 2 3 6 7 8]	<3 2 2 3 3 2>
6-z44A.0	[0 1 2 5 6 9]	<3 1 3 4 3 1>
6-z44B.0	[0 3 4 7 8 9]	<3 1 3 4 3 1>
6-z45.0	[0 2 3 4 6 9]	<2 3 4 2 2 2>
6-z46A.0	[0 1 2 4 6 9]	<2 3 3 3 3 1>
6-z46B.0	[0 3 5 7 8 9]	<2 3 3 3 3 1>
6-z47A.0	[0 1 2 4 7 9]	<2 3 3 2 4 1>
6-z47B.0	[0 2 5 7 8 9]	<2 3 3 2 4 1>

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
6-z48.0	[0 1 2 5 7 9]	<2 3 2 3 4 1>
6-z49.0	[0 1 3 5 6 9]	<2 2 4 3 2 2>
6-z50.0	[0 1 4 6 7 9]	<2 2 4 2 3 2>
7-1.0	[0 1 2 3 4 5 6]	<6 5 4 3 2 1>
7-2A.0	[0 1 2 3 4 5 7]	<5 5 4 3 3 1>
7-2B.0	[0 2 3 4 5 6 7]	<5 5 4 3 3 1>
7-3A.0	[0 1 2 3 4 5 8]	<5 4 4 4 3 1>
7-3B.0	[0 3 4 5 6 7 8]	<5 4 4 4 3 1>
7-4A.0	[0 1 2 3 4 6 7]	<5 4 4 3 3 2>
7-4B.0	[0 1 3 4 5 6 7]	<5 4 4 3 3 2>
7-5A.0	[0 1 2 3 5 6 7]	<5 4 3 3 4 2>
7-5B.0	[0 1 2 4 5 6 7]	<5 4 3 3 4 2>
7-6A.0	[0 1 2 3 4 7 8]	<5 3 3 4 4 2>
7-6B.0	[0 1 4 5 6 7 8]	<5 3 3 4 4 2>
7-7A.0	[0 1 2 3 6 7 8]	<5 3 2 3 5 3>
7-7B.0	[0 1 2 5 6 7 8]	<5 3 2 3 5 3>
7-8.0	[0 2 3 4 5 6 8]	<4 5 4 4 2 2>
7-9A.0	[0 1 2 3 4 6 8]	<4 5 3 4 3 2>
7-9B.0	[0 2 4 5 6 7 8]	<4 5 3 4 3 2>
7-10A.0	[0 1 2 3 4 6 9]	<4 4 5 3 3 2>
7-10B.0	[0 2 3 4 5 6 9]	<4 4 5 3 3 2>
7-11A.0	[0 1 3 4 5 6 8]	<4 4 4 4 4 1>
7-11B.0	[0 2 3 4 5 7 8]	<4 4 4 4 4 1>
7-z12.0	[0 1 2 3 4 7 9]	<4 4 4 3 4 2>
7-13A.0	[0 1 2 4 5 6 8]	<4 4 3 5 3 2>
7-13B.0	[0 2 3 4 6 7 8]	<4 4 3 5 3 2>
7-14A.0	[0 1 2 3 5 7 8]	<4 4 3 3 5 2>
7-14B.0	[0 1 3 5 6 7 8]	<4 4 3 3 5 2>
7-15.0	[0 1 2 4 6 7 8]	<4 4 2 4 4 3>
7-16A.0	[0 1 2 3 5 6 9]	<4 3 5 4 3 2>
7-16B.0	[0 1 3 4 5 6 9]	<4 3 5 4 3 2>
7-z17.0	[0 1 2 4 5 6 9]	<4 3 4 5 4 1>
7-z18A.0	[0 1 4 5 6 7 9]	<4 3 4 4 4 2>
7-z18B.0	[0 1 4 6 7 8 9]	<4 3 4 4 4 2>
7-19A.0	[0 1 2 3 6 7 9]	<4 3 4 3 4 3>
7-19B.0	[0 1 2 3 6 8 9]	<4 3 4 3 4 3>
7-20A.0	[0 1 2 5 6 7 9]	<4 3 3 4 5 2>
7-20B.0	[0 1 2 5 7 8 9]	<4 3 3 4 5 2>
7-21A.0	[0 1 2 4 5 8 9]	<4 2 4 6 4 1>
7-21B.0	[0 1 3 4 5 8 9]	<4 2 4 6 4 1>
7-22.0	[0 1 2 5 6 8 9]	<4 2 4 5 4 2>
7-23A.0	[0 2 3 4 5 7 9]	<3 5 4 3 5 1>
7-23B.0	[0 2 4 5 6 7 9]	<3 5 4 3 5 1>
7-24A.0	[0 1 2 3 5 7 9]	<3 5 3 4 4 2>
7-24B.0	[0 2 4 6 7 8 9]	<3 5 3 4 4 2>
7-25A.0	[0 2 3 4 6 7 9]	<3 4 5 3 4 2>
7-25B.0	[0 2 3 5 6 7 9]	<3 4 5 3 4 2>
7-26A.0	[0 1 3 4 5 7 9]	<3 4 4 5 3 2>
7-26B.0	[0 2 4 5 6 8 9]	<3 4 4 5 3 2>
7-27A.0	[0 1 2 4 5 7 9]	<3 4 4 4 5 1>
7-27B.0	[0 2 4 5 7 8 9]	<3 4 4 4 5 1>
7-28A.0	[0 1 3 5 6 7 9]	<3 4 4 4 3 3>
7-28B.0	[0 2 3 4 6 8 9]	<3 4 4 4 3 3>
7-29A.0	[0 1 2 4 6 7 9]	<3 4 4 3 5 2>
7-29B.0	[0 2 3 5 7 8 9]	<3 4 4 3 5 2>
7-30A.0	[0 1 2 4 6 8 9]	<3 4 3 5 4 2>
7-30B.0	[0 1 3 5 7 8 9]	<3 4 3 5 4 2>

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
7-31A.0	[0 1 3 4 6 7 9]	<3 3 6 3 3 3>
7-31B.0	[0 2 3 5 6 8 9]	<3 3 6 3 3 3>
7-32A.0	[0 1 3 4 6 8 9]	<3 3 5 4 4 2>
7-32B.0	[0 1 3 5 6 8 9]	<3 3 5 4 4 2>
7-33.0	[0 1 2 4 6 8 10]	<2 6 2 6 2 3>
7-34.0	[0 1 3 4 6 8 10]	<2 5 4 4 4 2>
7-35.0	[0 1 3 5 6 8 10]	<2 5 4 3 6 1>
7-z36A.0	[0 1 2 3 5 6 8]	<4 4 4 3 4 2>
7-z36B.0	[0 2 3 5 6 7 8]	<4 4 4 3 4 2>
7-z37.0	[0 1 3 4 5 7 8]	<4 3 4 5 4 1>
7-z38A.0	[0 1 2 4 5 7 8]	<4 3 4 4 4 2>
7-z38B.0	[0 1 3 4 6 7 8]	<4 3 4 4 4 2>
8-1.0	[0 1 2 3 4 5 6 7]	<7 6 5 4 4 2>
8-2A.0	[0 1 2 3 4 5 6 8]	<6 6 5 5 4 2>
8-2B.0	[0 2 3 4 5 6 7 8]	<6 6 5 5 4 2>
8-3.0	[0 1 2 3 4 5 6 9]	<6 5 6 5 4 2>
8-4A.0	[0 1 2 3 4 5 7 8]	<6 5 5 5 5 2>
8-4B.0	[0 1 3 4 5 6 7 8]	<6 5 5 5 5 2>
8-5A.0	[0 1 2 3 4 6 7 8]	<6 5 4 5 5 3>
8-5B.0	[0 1 2 4 5 6 7 8]	<6 5 4 5 5 3>
8-6.0	[0 1 2 3 5 6 7 8]	<6 5 4 4 6 3>
8-7.0	[0 1 2 3 4 5 8 9]	<6 4 5 6 5 2>
8-8.0	[0 1 2 3 4 7 8 9]	<6 4 4 5 6 3>
8-9.0	[0 1 2 3 6 7 8 9]	<6 4 4 4 6 4>
8-10.0	[0 2 3 4 5 6 7 9]	<5 6 6 4 5 2>
8-11A.0	[0 1 2 3 4 5 7 9]	<5 6 5 5 5 2>
8-11B.0	[0 2 4 5 6 7 8 9]	<5 6 5 5 5 2>
8-12A.0	[0 1 3 4 5 6 7 9]	<5 5 6 5 4 3>
8-12B.0	[0 2 3 4 5 6 8 9]	<5 5 6 5 4 3>
8-13A.0	[0 1 2 3 4 6 7 9]	<5 5 6 4 5 3>
8-13B.0	[0 2 3 5 6 7 8 9]	<5 5 6 4 5 3>
8-14A.0	[0 1 2 4 5 6 7 9]	<5 5 5 5 6 2>
8-14B.0	[0 2 3 4 5 7 8 9]	<5 5 5 5 6 2>
8-z15A.0	[0 1 2 3 4 6 8 9]	<5 5 5 5 5 3>
8-z15B.0	[0 1 3 5 6 7 8 9]	<5 5 5 5 5 3>
8-16A.0	[0 1 2 3 5 7 8 9]	<5 5 4 5 6 3>
8-16B.0	[0 1 2 4 6 7 8 9]	<5 5 4 5 6 3>
8-17.0	[0 1 3 4 5 6 8 9]	<5 4 6 6 5 2>
8-18A.0	[0 1 2 3 5 6 8 9]	<5 4 6 5 5 3>
8-18B.0	[0 1 3 4 6 7 8 9]	<5 4 6 5 5 3>
8-19A.0	[0 1 2 4 5 6 8 9]	<5 4 5 7 5 2>
8-19B.0	[0 1 3 4 5 7 8 9]	<5 4 5 7 5 2>
8-20.0	[0 1 2 4 5 7 8 9]	<5 4 5 6 6 2>
8-21.0	[0 1 2 3 4 6 8 10]	<4 7 4 6 4 3>
8-22A.0	[0 1 2 3 5 6 8 10]	<4 6 5 5 6 2>
8-22B.0	[0 1 2 3 5 7 9 10]	<4 6 5 5 6 2>
8-23.0	[0 1 2 3 5 7 8 10]	<4 6 5 4 7 2>
8-24.0	[0 1 2 4 5 6 8 10]	<4 6 4 7 4 3>
8-25.0	[0 1 2 4 6 7 8 10]	<4 6 4 6 4 4>
8-26.0	[0 1 3 4 5 7 8 10]	<4 5 6 5 6 2>
8-27A.0	[0 1 2 4 5 7 8 10]	<4 5 6 5 5 3>
8-27B.0	[0 1 2 4 6 7 9 10]	<4 5 6 5 5 3>
8-28.0	[0 1 3 4 6 7 9 10]	<4 4 8 4 4 4>
8-z29A.0	[0 1 2 3 5 6 7 9]	<5 5 5 5 5 3>
8-z29B.0	[0 2 3 4 6 7 8 9]	<5 5 5 5 5 3>
9-1.0	[0 1 2 3 4 5 6 7 8]	<8 7 6 6 6 3>
9-2A.0	[0 1 2 3 4 5 6 7 9]	<7 7 7 6 6 3>

Forte ID (.0)	Pitch classes	Interval vector <1 2 3 4 5 6>
9-2B.0	[0 2 3 4 5 6 7 8 9]	<7 7 7 6 6 3>
9-3A.0	[0 1 2 3 4 5 6 8 9]	<7 6 7 7 6 3>
9-3B.0	[0 1 3 4 5 6 7 8 9]	<7 6 7 7 6 3>
9-4A.0	[0 1 2 3 4 5 7 8 9]	<7 6 6 7 7 3>
9-4B.0	[0 1 2 4 5 6 7 8 9]	<7 6 6 7 7 3>
9-5A.0	[0 1 2 3 4 6 7 8 9]	<7 6 6 6 7 4>
9-5B.0	[0 1 2 3 5 6 7 8 9]	<7 6 6 6 7 4>
9-6.0	[0 1 2 3 4 5 6 8 10]	<6 8 6 7 6 3>
9-7A.0	[0 1 2 3 4 5 7 8 10]	<6 7 7 6 7 3>
9-7B.0	[0 1 2 3 4 5 7 9 10]	<6 7 7 6 7 3>
9-8A.0	[0 1 2 3 4 6 7 8 10]	<6 7 6 7 6 4>
9-8B.0	[0 1 2 3 4 6 8 9 10]	<6 7 6 7 6 4>
9-9.0	[0 1 2 3 5 6 7 8 10]	<6 7 6 6 8 3>
9-10.0	[0 1 2 3 4 6 7 9 10]	<6 6 8 6 6 4>
9-11A.0	[0 1 2 3 5 6 7 9 10]	<6 6 7 7 7 3>
9-11B.0	[0 1 2 3 5 6 8 9 10]	<6 6 7 7 7 3>
9-12.0	[0 1 2 4 5 6 8 9 10]	<6 6 6 9 6 3>
10-1.0	[0 1 2 3 4 5 6 7 8 9]	<9 8 8 8 8 4>
10-2.0	[0 1 2 3 4 5 6 7 8 10]	<8 9 8 8 8 4>
10-3.0	[0 1 2 3 4 5 6 7 9 10]	<8 8 9 8 8 4>
10-4.0	[0 1 2 3 4 5 6 8 9 10]	<8 8 8 9 8 4>
10-5.0	[0 1 2 3 4 5 7 8 9 10]	<8 8 8 8 9 4>
10-6.0	[0 1 2 3 4 6 7 8 9 10]	<8 8 8 8 8 5>
11-1.0	[0 1 2 3 4 5 6 7 8 9 10]	<10 10 10 10 10 5>
12-1.0	[0 1 2 3 4 5 6 7 8 9 10 11]	<12 12 12 12 12 6>